

## SELMA 13 WNW, AL, USA

WMO: 722360

Lat: 32.4567N

Lon: 87.2422W

Elev: 193

StdP: 14.59

Time Zone: -6.00 (NAC)

Period: 05-19

WBAN: 63858

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|------|------|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 21.0       | 24.7 | 11.0                          | 9.7 | 27.1 | 15.3 | 12.0 | 31.3 | 11.8                  | 46.8 | 10.8 | 46.1 | 1.8                   | N/A  | 0.321 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 19.4                   | 95.6            | 76.1 | 93.4 | 76.3 | 91.4 | 76.0 | 79.6                | 89.7 | 78.8 | 88.6 | 78.1 | 87.6 | 3.7                  | N/A  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 77.1                            | 141.9 | 84.5 | 76.2 | 137.7 | 83.8 | 75.3 | 133.7 | 83.1 | 43.2          | 89.9 | 42.4 | 88.8 | 41.6 | 87.6 | 83.1              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |     |                    |      | n-Year Return Period Values of Extreme Temperature |     |            |       |            |       |            |       |     |       |
|-------------------|------|-----|----------------------------|-----|--------------------|------|----------------------------------------------------|-----|------------|-------|------------|-------|------------|-------|-----|-------|
|                   |      |     | Mean                       |     | Standard Deviation |      | n=5 years                                          |     | n=10 years |       | n=20 years |       | n=50 years |       |     |       |
| 1%                | 2.5% | 5%  | Min                        | Max | Min                | Max  | Min                                                | Max | Min        | Max   | Min        | Max   | Min        | Max   |     |       |
| (a)               | (b)  | (c) | (e)                        | (f) | (g)                | (h)  | (i)                                                | (j) | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |     |       |
| (4)               | 9.7  | 8.4 | 7.2                        | DB  | 14.5               | 98.0 | 4.0                                                | 2.9 | 11.7       | 100.0 | 9.3        | 101.7 | 7.1        | 103.4 | 4.2 | 105.5 |
| (5)               |      |     |                            | WB  | 13.5               | 81.5 | 4.7                                                | 0.7 | 10.2       | 82.0  | 7.4        | 82.4  | 4.8        | 82.8  | 1.4 | 83.3  |

## Monthly Climatic Design Conditions

|                                                                               |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|-------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                               |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                           | DBAvg       | 64.0   | 45.2  | 49.7  | 56.7  | 63.6  | 70.8  | 78.4  | 80.3  | 80.4  | 75.7  | 64.4  | 53.2  | 48.8 |
|                                                                               | DBStd       | 14.56  | 10.52 | 10.10 | 8.76  | 6.68  | 5.72  | 3.26  | 2.75  | 3.46  | 4.89  | 8.38  | 8.60  | 9.83 |
|                                                                               | HDD50       | 603    | 223   | 123   | 38    | 3     | 0     | 0     | 0     | 0     | 0     | 4     | 62    | 150  |
|                                                                               | HDD65       | 2450   | 616   | 437   | 278   | 105   | 23    | 0     | 0     | 0     | 2     | 117   | 364   | 508  |
|                                                                               | CDD50       | 5716   | 75    | 114   | 246   | 410   | 646   | 852   | 939   | 944   | 770   | 449   | 158   | 113  |
|                                                                               | CDD65       | 2088   | 2     | 9     | 21    | 62    | 203   | 402   | 474   | 479   | 322   | 97    | 11    | 6    |
|                                                                               | CDH74       | 19485  | 7     | 38    | 173   | 590   | 1873  | 3879  | 4483  | 4545  | 2847  | 955   | 76    | 19   |
|                                                                               | CDH80       | 8383   | 0     | 1     | 14    | 88    | 633   | 1819  | 2095  | 2219  | 1227  | 280   | 7     | 0    |
| Wind                                                                          | WSAvg       | 2.6    | 3.6   | 3.8   | 3.6   | 2.7   | 2.1   | 2.0   | 1.9   | 2.1   | 2.0   | 2.2   | 2.6   | 3.2  |
| Precipitation                                                                 | PrecAvg     | 52.0   | 4.9   | 5.0   | 5.7   | 4.2   | 3.7   | 4.2   | 5.0   | 4.1   | 3.7   | 2.8   | 3.8   | 4.8  |
|                                                                               | PrecMax     | 67.6   | 9.1   | 8.0   | 11.7  | 10.3  | 8.9   | 9.4   | 11.7  | 10.5  | 10.2  | 8.7   | 8.2   | 10.6 |
|                                                                               | PrecMin     | 27.5   | 1.9   | 1.7   | 1.1   | 1.3   | 0.2   | 1.1   | 1.0   | 1.2   | 0.6   | 0.2   | 0.7   | 2.4  |
|                                                                               | PrecStd     | 9.2    | 2.1   | 1.8   | 3.0   | 2.1   | 2.4   | 2.4   | 2.5   | 2.2   | 2.2   | 1.9   | 2.0   | 1.9  |
| Monthly Design<br>Dry Bulb and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 74.8  | 78.3  | 82.1  | 84.7  | 91.2  | 96.6  | 96.4  | 100.0 | 95.1  | 90.1  | 80.5  | 76.8 |
|                                                                               |             | MCWB   | 64.8  | 67.4  | 65.6  | 69.9  | 73.1  | 75.6  | 77.5  | 75.8  | 73.5  | 69.7  | 66.6  | 67.2 |
|                                                                               | 2%          | DB     | 70.8  | 74.3  | 78.8  | 82.6  | 88.5  | 94.1  | 94.1  | 96.3  | 92.3  | 86.2  | 76.0  | 72.7 |
|                                                                               |             | MCWB   | 62.9  | 65.1  | 65.5  | 68.2  | 73.0  | 76.5  | 77.8  | 75.7  | 74.1  | 69.8  | 64.6  | 66.0 |
|                                                                               | 5%          | DB     | 66.6  | 70.5  | 75.5  | 80.4  | 86.1  | 91.8  | 92.1  | 93.5  | 89.9  | 83.2  | 72.6  | 69.3 |
|                                                                               |             | MCWB   | 61.1  | 63.1  | 63.9  | 67.1  | 72.1  | 76.3  | 77.2  | 76.4  | 74.1  | 68.2  | 63.4  | 63.8 |
|                                                                               | 10%         | DB     | 62.6  | 66.7  | 72.1  | 77.7  | 84.0  | 89.5  | 90.1  | 90.7  | 87.4  | 79.7  | 69.3  | 65.7 |
|                                                                               |             | MCWB   | 57.2  | 60.5  | 62.0  | 65.7  | 71.2  | 75.9  | 76.9  | 76.4  | 74.1  | 67.6  | 61.2  | 61.4 |
| Monthly Design<br>Wet Bulb and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 67.4  | 68.9  | 69.1  | 73.3  | 76.3  | 80.2  | 80.8  | 80.5  | 78.7  | 76.1  | 71.0  | 70.9 |
|                                                                               |             | MCDB   | 72.0  | 76.1  | 75.7  | 81.3  | 85.0  | 90.2  | 91.3  | 91.1  | 87.4  | 82.6  | 74.0  | 74.4 |
|                                                                               | 2%          | WB     | 64.8  | 67.1  | 67.5  | 71.2  | 74.7  | 79.1  | 79.6  | 79.5  | 77.4  | 74.5  | 68.5  | 68.0 |
|                                                                               |             | MCDB   | 68.9  | 73.5  | 75.4  | 78.8  | 83.9  | 89.2  | 89.8  | 89.5  | 86.5  | 81.3  | 72.5  | 71.8 |
|                                                                               | 5%          | WB     | 62.1  | 65.2  | 65.8  | 69.4  | 73.5  | 78.1  | 78.7  | 78.6  | 76.4  | 72.8  | 66.1  | 64.8 |
|                                                                               |             | MCDB   | 65.9  | 70.2  | 72.9  | 76.6  | 83.0  | 88.1  | 88.5  | 88.1  | 85.4  | 79.3  | 70.6  | 67.9 |
|                                                                               | 10%         | WB     | 58.4  | 62.2  | 63.9  | 67.8  | 72.6  | 77.1  | 78.0  | 77.8  | 75.2  | 70.2  | 62.4  | 61.7 |
|                                                                               |             | MCDB   | 61.7  | 66.4  | 70.6  | 74.7  | 81.6  | 86.7  | 87.5  | 86.9  | 84.0  | 76.2  | 67.7  | 65.6 |
| Mean Daily<br>Temperature<br>Range                                            | 5% DB       | MDBR   | 21.9  | 22.7  | 23.2  | 23.7  | 22.6  | 21.1  | 19.4  | 20.1  | 21.8  | 25.3  | 26.2  | 21.5 |
|                                                                               |             | MCDBR  | 23.6  | 24.6  | 24.9  | 25.5  | 25.5  | 24.7  | 22.5  | 24.7  | 24.8  | 27.4  | 26.8  | 23.1 |
|                                                                               |             | MCWBR  | 16.6  | 14.4  | 13.0  | 12.9  | 12.1  | 9.7   | 8.6   | 8.4   | 9.8   | 12.9  | 16.5  | 17.4 |
|                                                                               | 5% WB       | MCDBR  | 19.2  | 18.9  | 20.3  | 19.9  | 21.1  | 21.2  | 19.3  | 19.5  | 19.4  | 19.5  | 19.8  | 19.4 |
|                                                                               |             | MCWBR  | 15.1  | 12.9  | 12.0  | 11.5  | 10.8  | 9.5   | 8.5   | 8.2   | 8.8   | 10.6  | 15.0  | 16.1 |
| Clear-Sky<br>Solar<br>Irradiance                                              | taub        | 0.323  | 0.334 | 0.361 | 0.399 | 0.451 | 0.483 | 0.545 | 0.536 | 0.447 | 0.374 | 0.346 | 0.325 |      |
|                                                                               | taud        | 2.525  | 2.498 | 2.429 | 2.308 | 2.214 | 2.166 | 2.005 | 2.025 | 2.260 | 2.448 | 2.468 | 2.533 |      |
|                                                                               | Ebn at Noon | 285    | 293   | 291   | 283   | 268   | 258   | 241   | 242   | 260   | 275   | 273   | 278   |      |
|                                                                               | Edh at Noon | 27     | 30    | 35    | 41    | 46    | 48    | 56    | 54    | 41    | 32    | 28    | 25    |      |
| All-Sky Solar<br>Radiation                                                    | RadAvg      | 874    | 1055  | 1417  | 1778  | 1964  | 1988  | 1913  | 1772  | 1574  | 1325  | 991   | 751   |      |
|                                                                               | RadStd      | 64     | 134   | 131   | 132   | 134   | 154   | 127   | 104   | 126   | 159   | 97    | 78    |      |

## Historical Trends

|      |                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------|-------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|      |                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|      |                         | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| (46) | Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) | Regional (36 neighbors) | +0.88 | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | +266  | +176  |

Nomenclature: See separate page